ReliefLink: Extended Master Live Demonstration Script

Comprehensive, Long-Form Screen-by-Screen Technical Walkthrough for Mobile & Web Applications

Target Presentation Duration: 15 – 20 Minutes (Full Deep Dive)

Format: Paired Presenters (Mobile Lead & Web/Blockchain Lead)

Mobile Platform: Flutter / Dart (iOS, Android, Web Engine)

Web Portal: React 18, Vite, TailwindCSS, Chart.js, Axios

Payment Integration: PayMongo Automated API (QR Ph, GCash, E-wallets)

Consortium Blockchain: Hyperledger Besu (Private Ethereum PoA/QBFT)

PART A: EXTENDED MOBILE DONOR APPLICATION DEMONSTRATION

Scene 1: Application Launch, Cold Boot & Splash Initialization

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Pre-condition: App launched from clean state.
Step 1: Launch Flutter mobile app.
Step 2: Display the animated splash screen featuring the ReliefLink emblem and tagline 'Parish Relief & Donation System'.
Step 3: Point out the smooth transition as the app verifies local token validity via SharedPreferences and navigates forward.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Good day, members of the evaluation panel. We begin our live demonstration with the ReliefLink Mobile Application, engineered in Flutter for high-performance cross-platform deployment on Android, iOS, and mobile web browsers.

Upon cold boot, the application initializes our design system, establishes encryption salts, and checks local device storage for existing cryptographic sessions. If an authenticated session is detected, the donor is routed directly to their dashboard; otherwise, the app presents our secure authentication gateway with zero UI stuttering."

Scene 2: Donor Authentication, Form Validation & Session Persistence

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Login Screen (login_screen.dart).
Step 1: Intentionally leave the password blank and tap 'Login' to show instant form validation error alerts.
Step 2: Click the eye icon to toggle password visibility.
Step 3: Type email ikathpie3@gmail.com and valid password. Tap 'Login'.
Step 4: Highlight the smooth transition as JWT tokens are received from the Node.js backend and saved locally.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Security is paramount in financial platforms. Our authentication module implements client-side regex validation preventing malformed credentials from hitting our API servers. When the donor logs in, the backend issues an industry-standard JSON Web Token containing encrypted claims. This token is securely stored and attached as a Bearer authorization header to every subsequent HTTPS request. This ensures all transaction receipts and personal giving impact records remain confidential and uniquely bound to this parishioner."

Scene 3: Personalized Home Dashboard & Elastic Momentum Physics

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Home Screen (Tab 0).
Step 1: Observe the time-aware dynamic greeting banner ('Good morning/afternoon, Marie Jana Arillo').
Step 2: Actively drag and scroll the page up and down using the trackpad/mouse to demonstrate our custom AppScrollBehavior with BouncingScrollPhysics and multi-pointer drag devices.
Step 3: Point out the two prominent KPI cards: **TOTAL DONATED: ■160.00** and **TRANSACTIONS: 7**.
Step 4: Show the 'Support Sto. Domingo Church' Hero card and the 'Explore ReliefLink' quick navigation tiles.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Upon entering the Home Dashboard, parishioners are welcomed with a personalized, time-sensitive greeting and an immediate summary of their charitable contributions. Notice our KPI cards: this donor has contributed ■160.00 across 7 separate verified transactions.

From an engineering standpoint, observe the scroll behavior. By default, Flutter web disables mouse dragging and uses rigid clamping. We engineered a custom AppScrollBehavior combining bouncing elastic physics with touch, mouse, and trackpad drag support. This ensures whether a donor uses a smartphone touchscreen or a laptop trackpad, scrolling feels natural, responsive, and buttery smooth.

Notice that following our adviser's guidance, we kept the Home Screen clean and focused on quick actions, reserving deep financial audits for our specialized transparency module."

Scene 4: Destination Earmarking & Flexible Preset Amounts (Donate Tab)

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Bottom Navigation Bar → Tap the centered blue **Donate** button.
Step 1: Show the prominent, elevated floating action button design in the bottom navbar.
Step 2: Open the Destination Selector dropdown. Show available ministries: *Parish General Fund, LGBTQ, Elderly, PDL (Persons Deprived of Liberty), Urban Poor, Migrant Workers, Student Scholarships, Drug Rehabilitation*.
Step 3: Select **'Parish General Fund'**.
Step 4: Tap the preset chips: ■50, ■100, ■200, ■500, or type a custom amount of **■25.00** into the input box.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Now, let us examine the donation flow. On traditional giving portals, donations fall into a single opaque pool. In ReliefLink, donors possess granular agency. By tapping our central Donate button, the parishioner can earmark their gift for specific church apostolates—including Student Scholarships, Healthcare for the Elderly, or Calamity Relief.

Donors can tap preset amount chips or type custom amounts down to the exact peso. Our reactive state architecture recalculates totals instantly, ensuring complete transparency before a single centavo leaves the donor's account."

Scene 5: Automated PayMongo Cashless Checkout & Webhook Settlement

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Donation Confirmation Sheet.
Step 1: Tap **'Proceed to PayMongo Checkout'**.
Step 2: Display the secure PayMongo checkout interface showing Philippine payment rails: QR Ph, GCash, Maya, and Cards.
Step 3: Explain the automated fee pass-through and secure signature validation.
Step 4: Complete the simulated transaction and observe the return to the app with immediate confirmation.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

*"To eliminate the physical vulnerabilities of cash handling and manual bank transfer screenshot fraud, ReliefLink integrates **PayMongo**. When the donor confirms, our Express backend generates an authorized PayMongo checkout session supporting Philippine national standard QR Ph and GCash.*

The moment payment clears, PayMongo's secure webhook dispatches an encrypted payload to our backend server. Our server validates the cryptographic webhook signature, updates the donation state to verified, and triggers smart contract mining on the blockchain—all in under two seconds without requiring the donor to upload proof of payment."

Scene 6: Real-Time 5-Second Background Polling & Digital Receipts

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Bottom Navigation Bar → Tap **History** (Donation Summary).
Step 1: Show the list of transactions.
Step 2: Explain that a background Timer.periodic executes every 5 seconds calling load(showSpinner: false), updating data silently without showing full-screen loading spinners or resetting scroll position.
Step 3: Tap on the top donation card (■25.00) to expand the itemized digital receipt.
Step 4: Point out the payment channel (PayMongo QR Ph), Reference code (cs_fad...), and Date timestamp.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Navigating to the History tab reveals our real-time synchronization engine. Many mobile apps force users to manually swipe down to refresh. In ReliefLink, we implemented a periodic 5-second polling timer. Crucially, we designed this to execute silently in the background: if new disbursements occur or admin verifications complete, the data updates fluidly on screen without annoying loading spinners or jarring scroll resets.

Expanding any card reveals an official digital receipt complete with PayMongo reference identifiers, timestamps, and destination earmarks."

Scene 7: Hyperledger Besu Cryptographic Hash & Tamper-Proof Verification

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Expanded Receipt Card in History tab.
Step 1: Scroll inside the receipt to the blue **Blockchain Information** container at the bottom.
Step 2: Point to the monospace hexadecimal hash starting with 0x....
Step 3: Click the **Copy** icon button on the right.
Step 4: Show the green SnackBar notification: *'Blockchain hash copied to clipboard!'*.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

*"Now, observe the most critical technical innovation on this receipt: the **Blockchain Information** box. This long alphanumeric string starting with 0x is not an internal database ID—it is the cryptographic transaction hash generated when this donation was mined into a block on our private **Hyperledger Besu Ethereum blockchain**.*

By tapping the copy button, the donor copies this hash. This cryptographic receipt is mathematically permanent: neither parish priests, system administrators, nor external attackers can alter the amount, the timestamp, or the donor identity without invalidating the entire blockchain ledger."

Scene 8: Reports & Transparency — Sticky Nested Header Architecture

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Open **Reports & Transparency** from Home or drawer.
Step 1: Demonstrate the NestedScrollView with pinned SliverPersistentHeader.
Step 2: Scroll down: observe how the blue overview card (■26,923.00 Parish Outflow) smoothly slides up out of view.
Step 3: Observe how the 3 tabs (**My Contributions**, **Parish Expenses**, **Need Allocation (%)**) pin securely at the top.
Step 4: Scroll back up to show the overview card returning smoothly.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

*"Next, we enter our flagship transparency module: **Reports & Transparency**.*

Previously, fixed headers occupied over 50% of mobile screens, creating a claustrophobic scrolling experience. We completely re-architected this screen using Flutter's NestedScrollView and sliver persistent header delegates. As the donor scrolls, the top overview banner glides out of the viewport, while the tab navigation bar pins neatly beneath the app bar. This gives the donor 100% of their screen real estate to examine detailed audit records comfortably."

Scene 9: 'My Contributions' Utilization Gauge & Giving Impact Breakdown

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: First Tab of Transparency Screen (**'My Contributions'**).
Step 1: Point out the Giving Utilization Summary card: **Total Verified Given: ■160.00**.
Step 2: Point to the two-tone progress bar: **Disbursed to Relief Operations: ■45.00 (28%)** vs **Cash on Hand in Treasury: ■115.00 (72%)**.
Step 3: Scroll to the donation card for ■25.00 (General Fund) marked **FULLY USED**.
Step 4: Expand the card to show **GIVING IMPACT BREAKDOWN**: point out the event *'Community Kitchen Feeding Program'*, Disbursed Amount: ■*25.00*, Event Date: *10/14/2026*, Notes: *'Purchased warm nutritional meals for 15 children at the parish outreach center'*, and Balance remaining after event: ■*0.00*.
Step 5: Show the second card for ■30.00 marked **PARTIALLY USED** (■20.00 disbursed for Typhoon Relief, ■10.00 remaining cash on hand).

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

*"Here under the **'My Contributions'** tab, ReliefLink delivers what no other church donation system provides: real-time, post-donation utilization tracking.*

Notice our dynamic utilization gauge: it shows that out of ■160.00 given by this donor, ■45.00 or 28% has been actively deployed into community missions, while ■115.00 or 72% remains safely guarded in the parish treasury.

*Even more powerful is the **Giving Impact Breakdown**. When we expand this ■25.00 donation, the donor does not see a vague aggregate. They see the exact outreach program their money financed: ■25.00 allocated on October 14 for the 'Community Kitchen Feeding Program', with field notes confirming warm meals were provided to 15 children. Beside it, a ■30.00 donation shows 'Partially Used': ■20.00 deployed for Typhoon Relief, with ■10.00 remaining cash on hand. Every single peso is accounted for from deposit to community deployment."*

Scene 10: Parish Bulletins Filter & Account Security Settings

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Bottom Navbar → Tap **Announcements** (Tab 1), then tap **Profile** (Tab 4).
Step 1: On Announcements, click the category dropdown and filter between *All, General, Urgent Relief, Events*.
Step 2: On Profile Screen, show user personal details, tap 'Edit Profile', tap 'Change Password', and tap 'About ReliefLink'.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Concluding the mobile client: Module M7 keeps parishioners engaged with church life through categorized announcements and urgent disaster relief callouts. Module M8 gives donors complete sovereignty over their accounts—enabling profile updates, password rotations with rigorous policy validation, and access to parish system credentials."

PART B: EXTENDED WEB ADMINISTRATIVE & PRESCRIPTIVE PORTAL

Scene 11: Superadmin Authentication & Role-Based Access Control

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Switch to Chrome browser displaying the React Web Portal (localhost:3000).
Step 1: Enter Superadmin credentials (Francis Arillo).
Step 2: Click **'Sign In'**.
Step 3: Point out the Superadmin badge on the upper left sidebar and explain the RBAC permissions hierarchy (Superadmin, Admin, Staff, Viewer).

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Now, we transition to the Web Administrative Portal, engineered in React 18 and Vite for parish administrators, financial auditors, and clergy.

We authenticate using our Superadmin profile. ReliefLink enforces strict Role-Based Access Control: parish priests and head finance officers hold superadmin privileges, ministry volunteers hold staff clearance for attendance and relief distribution, while external auditors receive read-only verification rights."

Scene 12: Executive Financial Solvency Dashboard & Inflow/Outflow Trends

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Admin Dashboard Tab.
Step 1: Highlight the top metric strip: **Total Collected: ■161,343.00**, **Total Beneficiaries: 171**, **Actual Aid Disbursed: ■18,923.00**, and **Treasury Reserves: ■142,420.00**.
Step 2: Scroll through the interactive Chart.js visualizations showing monthly inflow vs outflow curves.
Step 3: Show the recent transactions audit table at the bottom.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"The Executive Dashboard serves as the central financial command center for Sto. Domingo Parish. At a single glance, the parish council monitors total treasury solvency: ■161,343 in gross community inflows, ■18,923 in disbursed community assistance, and ■142,420 in active liquid reserves across all ministries.

Our reactive Chart.js charts map donation seasonality and relief outflow, allowing parish leadership to anticipate budgetary needs before crisis events occur."

Scene 13: Donation Management Ledger & Multi-Status Filtering

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Sidebar → Click **Donation Management**.
Step 1: Show the full donation data table.
Step 2: Click through the filter tabs: **All**, **Pending**, **Approved**, **Cash on Hand**, **Partially Used**, **Fully Used**, and **Canceled**.
Step 3: Use the search bar to type donor name Marie or reference number cs_fad to show instant client-side filtering.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Module W3: Donation Management provides an exhaustive registry of every physical and digital contribution. Notice the granular status segmentation: administrators can filter between Pending collections, Approved Cash on Hand awaiting deployment, Partially Used funds, and Fully Disbursed records.

Crucially, canceled or rejected transactions are segregated, preventing non-cleared gifts from polluting treasury solvency metrics. Our high-speed search bar allows auditors to locate any record by donor name, email, or PayMongo checkout session ID within milliseconds."

Scene 14 & 15: Executing a Relief Fund Disbursement & Dynamic Balance Recalculation

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Donation Management Table → Click on verified donation (■30.00 General Fund).
Step 1: In the Donation Details Modal, click **Edit Status / Disburse**.
Step 2: Set status dropdown to: **'partially used'**.
Step 3: Enter Disbursed Amount: **20.00**.
Step 4: Enter Event Purpose: **'Typhoon Relief Operation'**.
Step 5: Enter Event Notes: **'Purchased 5 emergency food relief packs for families in Barangay Sto. Domingo'**.
Step 6: Click **Save Changes**. Point out the toast: *'Donation updated: PARTIALLY USED (■20.00 allocated to Typhoon Relief Operation)'*.
Step 7: Show the modal displaying the updated remaining balance of ■10.00 and the new entry in the usageLogs history.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Now, let us demonstrate how the parish deploys funds in practice. When disaster strikes or a feeding program is scheduled, the administrator opens the verified donation record and initiates a disbursement.

Here, we select 'Partially Used', enter ■20.00 for the Typhoon Relief Operation, and log official notes confirming that 5 emergency packs were purchased. The system executes our balance formula: original ■30.00 minus ■20.00 disbursed leaves exactly ■10.00 remaining cash on hand.

The instant we click Save, this record is committed to MongoDB and broadcast over our REST API. As we verified earlier on mobile, this exact ■20.00 entry appeared under the donor's 'My Contributions' impact breakdown in real time."

Scene 16 & 17: Beneficiary Demographic Registry & Scholar Grant Renewal Matrix

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Sidebar → Click **Beneficiary Management**.
Step 1: Show the active registry of 171 evaluated members across sectors (Scholars, Seniors/PWD, Indigent Families, Calamity Victims, Livelihood).
Step 2: Click the **Scholar Grant Renewal Matrix**.
Step 3: Point out the multi-criteria evaluation columns: Scholar Name, GPA / Academic Standing, Attendance Percentage, Community Service Hours, and Renewal Recommendation.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"Module W5 manages our registered community beneficiaries. Charity must be systematic: ReliefLink tracks 171 active members across verified vulnerability sectors.

*For student scholars, we built a specialized **Multi-Criteria Scholar Grant Renewal Matrix**. Rather than subjective reviews, the system evaluates semester GPA, church outreach attendance, and service hours to rank scholars objectively for tuition renewal."*

Scene 18: Prescriptive Decision Support Engine — Urgency Weights & Allocation Gaps

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Sidebar → Click **Reports**.
Step 1: Highlight the top 4 cards: **OPTIMIZATION BASELINE (171 Members, ■161,343 Pool)**, **PRESCRIBED AID TARGET (■161,343)**, **ACTUAL AID DISBURSED (■18,923)**, and **REMAINING ALLOCATION GAP (■142,420.00)**.
Step 2: Scroll through the table: highlight **Ministry Sector**, **Urgency Multiplier** (1.6x Scholars, 1.4x Indigent, 1.3x Seniors, 1.2x Calamity, 1.0x Livelihood), **Headcount**, **Target Share**, **Prescribed Target**, **Variance / Gap**, and **Prescriptive Directive**.
Step 3: Explain that 'Gap: ■30,222.00' represents underfunded sectors, and point to the blue **'Manage Aid'** button.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

*"Now we arrive at the intellectual core of our capstone research: **Module W6: The Prescriptive Decision Support Engine**.*

*In data science, descriptive analytics tells you what happened, predictive analytics forecasts what might happen, but **prescriptive analytics tells you what you should DO**.*

Our mathematical optimization model evaluates registered beneficiaries and weights them by societal urgency: a 1.6x multiplier for student scholars with non-negotiable school fees, 1.4x for indigent nutrition, and 1.3x for senior healthcare. It then calculates the optimal prescribed budget per sector against our ■161,343 treasury pool.

*Notice the **Remaining Allocation Gap**: ■142,420 remains to be disbursed. The system generates explicit Prescriptive Directives—such as: 'Disburse ■30,222.00 to reach optimal 100% allocation for scholars'. By clicking 'Manage Aid', the parish administrator is guided directly to distribute funds and eliminate the gap."*

Scene 19 & 20: Operational Expense Auditing, Document Archiving & Staff Governance

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Screen: Click through **Expenses**, **Documents**, **Announcement**, and **User Management**.
Step 1: Show uploaded expense receipts in Expenses.
Step 2: Show the financial audit PDF repository in Documents.
Step 3: Show the announcement publishing editor and the staff access management matrix.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

"The administrative ecosystem is rounded out by Module W7 for logging parish operational expenses with attached digital receipt proofs, Module W8 for centralizing audit certificates and voucher archives, Module W9 for broadcasting parish bulletins, and Module W10 for managing staff credentials and revoking inactive permissions."

PART C: BLOCKCHAIN ARCHITECTURE & MASTER PANEL Q&A; DEFENSE SHEET

Technical Deep Dive: Hyperledger Besu Consortium Blockchain Subsystem

■■ LIVE SCREEN DEMO ACTION & EXACT STEPS TO REPLICATE:

Visual Display: Terminal showing Besu node logs or VS Code displaying DonationRegistry.sol smart contract.
Action: Point out the Solidity event DonationRecorded, the function recordDonation(...), and the QBFT consensus mechanism.

■■ EXTENDED SPOKEN NARRATIVE (SPEAKER SCRIPT):

*"To guarantee absolute trust, our backend connects to a private **Hyperledger Besu Ethereum blockchain** running our Solidity smart contract DonationRegistry.sol.*

Unlike public mainnets that charge volatile cryptocurrency gas fees, Besu is an enterprise-grade consortium chain configured with zero gas price (`min-gas-price = 0`). The diocese and participating parishes serve as validator nodes without financial overhead.

Whenever a donation is approved, an immutable transaction block is minted. If a malicious insider with database access attempts to alter donor amounts or fabricate phantom expenses in MongoDB, our verification scripts immediately detect a hash mismatch with the blockchain, mathematically preventing fraud."

Master Panel Q&A; Defense Strategy Sheet (Top 6 Difficult Questions)

Anticipated Panel Question	Recommended High-Scoring Technical Defense Answer
1. Why use a blockchain instead of a secure relational database like PostgreSQL?	Traditional databases possess superusers (DBAs) with root permissions capable of executing silent UPDATE or DELETE queries to erase evidence. A blockchain is strictly append-only: once a block hash is computed and linked to prior headers, historical records cannot be altered without invalidating the entire chain, providing mathematically provable tamper evidence.
2. Do ordinary churchgoers need crypto knowledge or MetaMask?	No. Parishioners donate using standard Philippine payment channels (GCash, Maya, QR Ph via PayMongo). The blockchain works transparently in the background on private Hyperledger Besu nodes, exposing only the verifiable cryptographic transaction hash on their digital receipt.
3. What makes your Prescriptive Analytics 'prescriptive' rather than descriptive?	Descriptive analytics merely tells you how much money was collected. Our prescriptive engine evaluates active demographic registrations, applies calibrated urgency multipliers (1.6x scholars, 1.4x indigent feeding), computes the exact recommended budget per sector, and generates explicit prescriptive directives to close remaining allocation gaps.
4. How does the mobile app stay updated when an admin disburses funds?	We implemented a periodic silent background polling architecture in Flutter (5-second polling timer) that refreshes transaction usage logs and status badges without causing full-screen spinners or resetting scroll positions.
5. How do you prevent canceled or rejected donations from distorting reports?	Canceled or failed transactions are strictly filtered out in both the backend aggregation pipeline and the mobile 'My Contributions' view, ensuring only verified funds in the treasury are counted.
6. What is the business value of ReliefLink to Sto. Domingo Parish?	It restores community faith in church stewardship, eliminates the physical vulnerability of cash collection theft, reduces administrative bookkeeping overhead by over 70%, and ensures aid is distributed where community need is objectively highest.